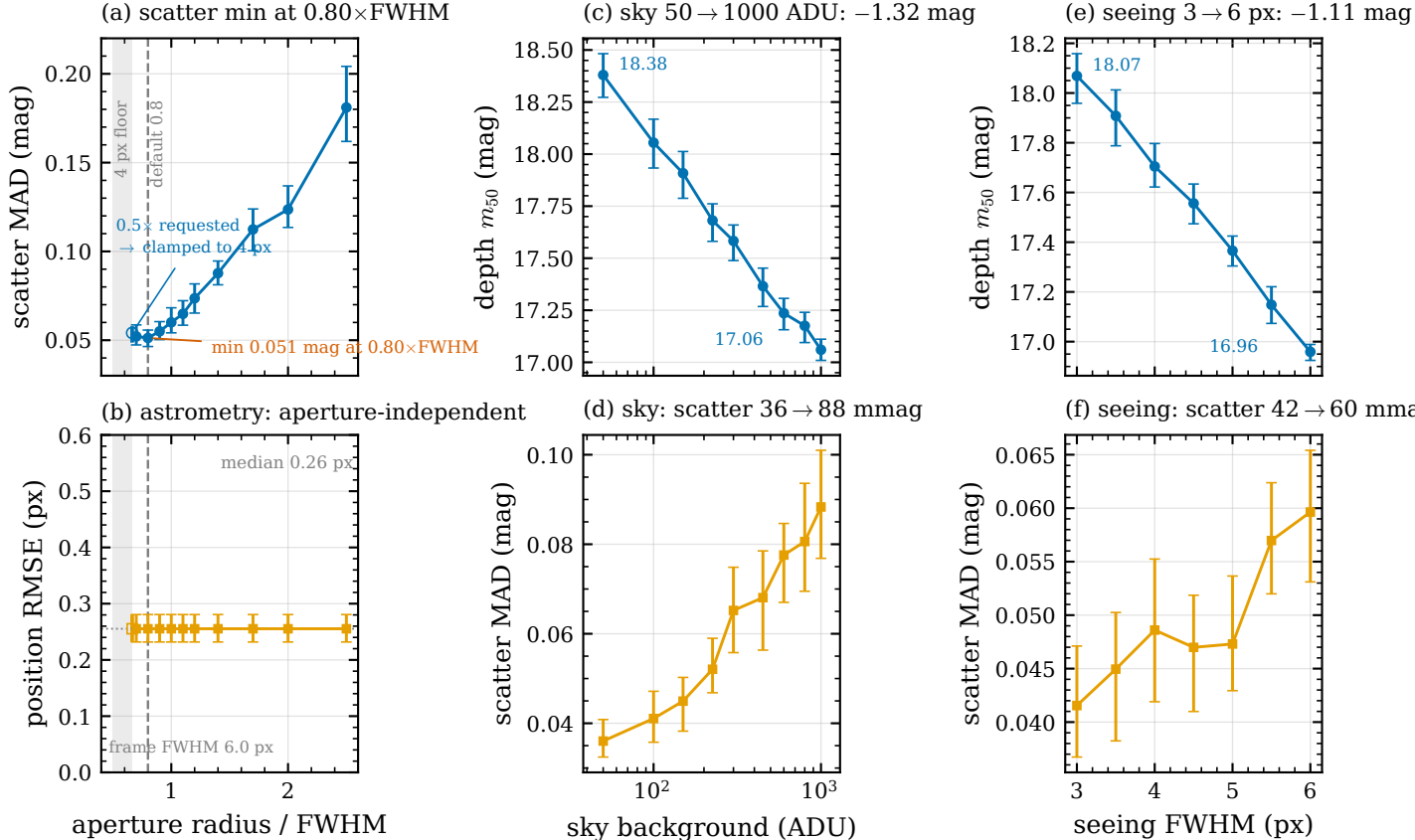




Synthetic frames: APEX generator (`apex.benchmark.synthetic_frame`), seed 20260702 · $1024^2 \text{ px} \cdot 435 \text{ field stars} \cdot \text{gain } 1.5 \text{ e}^-/\text{ADU} \cdot \text{RN } 5 \text{ e}^- \cdot \text{ZP } 25 \cdot r$. (a,b): one fixed frame, FWHM 6 px, sky 150 ADU; (c-f): frame regenerated per condition from the FWHM 3.5 px / sky 150 ADU baseline. Each point: $30 \text{ trials} \times 40 \text{ injected stars}$ (empirical-PSF injection $\rightarrow$ production Step-4 detection $\rightarrow$ forced aperture photometry). Bars: 95% CI, cluster bootstrap over trials; depth CI from the production completeness fit.